

CS301: Computer Networks — Protocols, TCP/IP, and HTTP/3 QUIC

1. Network Architecture and the Transport Layer

Computer networks operate across layered architectures, notably the 7-layer OSI model and the 4-layer TCP/IP protocol suite. At the transport layer, the Transmission Control Protocol (TCP) offers reliable, connection-oriented byte-stream delivery with sequence numbering, acknowledgment packets, and checksum verification. The User Datagram Protocol (UDP) provides connectionless, lightweight, best-effort datagram delivery without retransmission, ideal for real-time media streaming and online gaming.

2. The TCP 3-Way Handshake and Connection Teardown

To establish a reliable TCP connection, endpoints execute a 3-Way Handshake: 1) Client sends SYN packet with an initial sequence number (ISN_c); 2) Server responds with SYN-ACK packet containing server ISN_s and ACK = ISN_c + 1; 3) Client replies with ACK packet with ACK = ISN_s + 1. Connection teardown uses a 4-Way Handshake (FIN, ACK, FIN, ACK) with a TIME_WAIT state (usually 2MSL = 2 minutes) on the active closing side to ensure delayed segments in the network do not corrupt subsequent new connections.

3. TCP Congestion Control Algorithms

TCP manages network congestion using mechanisms defined by Van Jacobson: Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery. During Slow Start, the congestion window (cwnd) doubles every round-trip time (RTT) until reaching the slow start threshold (ssthresh). When packet loss is detected via 3 duplicate ACKs, Fast Retransmit immediately resends the missing segment without waiting for a retransmission timeout timer.

4. Evolution of Web Protocols: HTTP/1.1, HTTP/2, and HTTP/3

HTTP/1.1 introduced persistent connections and chunked transfer encoding, but suffered from Head-of-Line (HoL) blocking at the application level. HTTP/2 resolved this by introducing binary framing, header compression (HPACK), and multiplexed streams over a single TCP connection. However, TCP-level packet loss still stalls all multiplexed streams in HTTP/2. HTTP/3 eliminates TCP HoL blocking by running over QUIC, a transport protocol built on top of UDP. QUIC integrates TLS 1.3 encryption natively, reduces connection establishment latency to 0-RTT for repeated visits, and supports seamless connection migration across IP changes using Connection IDs.